



Pronob Chandra Sarkar

B.Sc. in Computer Science and Engineering Student

Professional Profile

Highly motivated and detail-oriented undergraduate student with a demonstrated expertise in **Machine Learning**, **XAI**, and Mobile Development. Seeking to leverage foundational CSE knowledge, strong academic performance (CGPA: **3.56**), and published research to secure a challenging role in research or software development.

Education

- 2022 – 2025 **B.Sc. in Computer Science and Engineering (CSE)**, *Daffodil International University (DIU)*, Savar, Dhaka, CGPA: **3.56** / 4.00
Currently completing coursework in Data Structures, Algorithms, Object-Oriented Programming, and Database Management.
- 2020 **Higher Secondary Certificate (HSC) Equivalent**, *Savar Govt. College*, Savar, Dhaka, GPA: **4.75** / 5.00
Completed all requirements with a focus on Science.
- 2018 **Secondary School Certificate (SSC)**, *Shukurjan-Zinnat-Ali-Adarsha-School*, Savar, Dhaka, GPA: **4.61** / 5.00
Foundation in core academic subjects.

Publications & Research

- 2025 **UDCAD-DFL-DL: A Unique Dataset for Classifying and Detecting Agricultural Diseases in Dragon Fruits and Leaves**, *Data in Brief (Elsevier)*
- Co-authored and published a peer-reviewed article focusing on the classification and detection of agricultural diseases using machine learning techniques.
 - Developed and structured a unique, publicly available dataset (UDCAD-DFL-DL) crucial for future research in computer vision and agricultural AI.
 - DOI: [10.1016/j.dib.2025.111411](https://doi.org/10.1016/j.dib.2025.111411)

Technical Skills

- Programming Languages C/C++, Java, Python (TensorFlow, Keras), Dart, JavaScript, SQL.
- ML Models & Techniques **Explainable AI (XAI)**, Transfer Learning, CNNs (VGG16, VGG19, ResNet, MobileNet), Decision Trees, Clustering (K-Means), Regression Models.
- Frameworks & Libraries Flutter (Mobile Development), HTML5, CSS3, Tailwind CSS, NumPy, Pandas, Scikit-learn, Matplotlib.
- Database/Tools MySQL/PostgreSQL (SQL), Database Design, Git/GitHub (Version Control), VS Code, Linux.

Savar, Dhaka, – Bangladesh

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Academic Projects

- November 2025 **Mobile Disease Classification App & XAI Integration**, *Academic Project*, DIU
- **Technologies:** Flutter, Dart, Python/Flask, Deep Learning Models, XAI Methods
 - **Role:** Lead Developer/Researcher
 - Developed a cross-platform mobile application using Flutter for real-time classification of agricultural diseases via image inputs (related to published research).
 - Integrated custom deep learning models (**VGG16**, **ResNet50**) and focused on implementing **Explainable AI (XAI)** methods to interpret model predictions.
 - Utilized Python/Flask for API deployment, ensuring robust real-time inference.
- November 2024 **Tournament Management System**, *Web Engineering Course*, DIU
- **Technologies:** HTML, CSS, JavaScript, Backend Scripting, SQL
 - **Role:** Full-Stack Developer
 - Designed and implemented a full-stack website to manage sports tournaments, handling user registration, scheduling, match results, and ranking systems.
 - Developed complex SQL queries and ensured efficient database handling for reliable, scalable data management.
 - Focused on user experience and robust data integrity, fulfilling the requirements for the web engineering course.
- June 2023 **Algorithm Visualizer Site**, *Algorithms Course*, DIU
- **Technologies:** JavaScript, HTML5, CSS3
 - **Role:** Core Developer
 - Created a dynamic web application to visually demonstrate key algorithms (e.g., Sorting, Graph Traversal), improving comprehension of complex data structures.
 - Implemented interactive UI components allowing users to input data and observe the step-by-step algorithm execution process.
- February 2024 **AI-Powered Tic-Tac-Toe Game**, *Artificial Intelligence Course*, DIU
- **Technologies:** Python, Minimax/Search Algorithms
 - **Role:** AI Implementation Specialist
 - Developed a perfect-play Tic-Tac-Toe game agent utilizing the Minimax algorithm and search tree optimization.
 - Focused on efficient game logic implementation to ensure instantaneous, optimal decision-making against human players.

Interests

Cycling, traveling, exploring new technologies, and contributing to open-source projects.